Chapter - 2 Market Economics

2.1 Market, Demand, Supply, Relationship between demand and supply and Market Equilibrium

2.2 Elasticity , Application of elasticity and Government Policies (e.g., price ceilings and price floors) **(Numerical)**

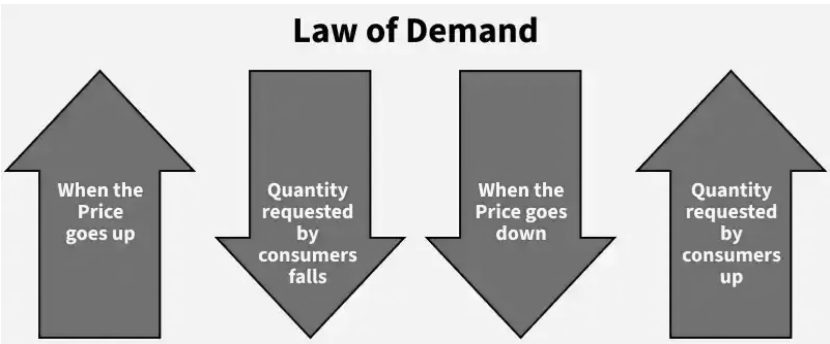
2.3 Externalities , Market inefficiency and Market Failure

2.4 Firm Behavior

2.1 Market, Demand, Supply, Relationship between demand and supply and Market Equilibrium

Demand in economics is the consumer's desire and ability to purchase a goods or service. Demand in economics is how many goods and services are bought at various prices during a certain period of time. "Every demand or want supported by the willingness and ability to buy" constitutes demand for a particular product or service.".

Law of Demand : "All other things being equal, demand varies inversely with price".



NOTE: The relationship between price and quantity is inverse.

Factors Influencing Demand
The shape of the demand curve is influenced by the following factors;

i) Income of the people
ii) Taste of consumers
iii) Prices of related goods
iv) Custom and fashions
v) Weather
vi) Future expectations
vii) Price of commodity

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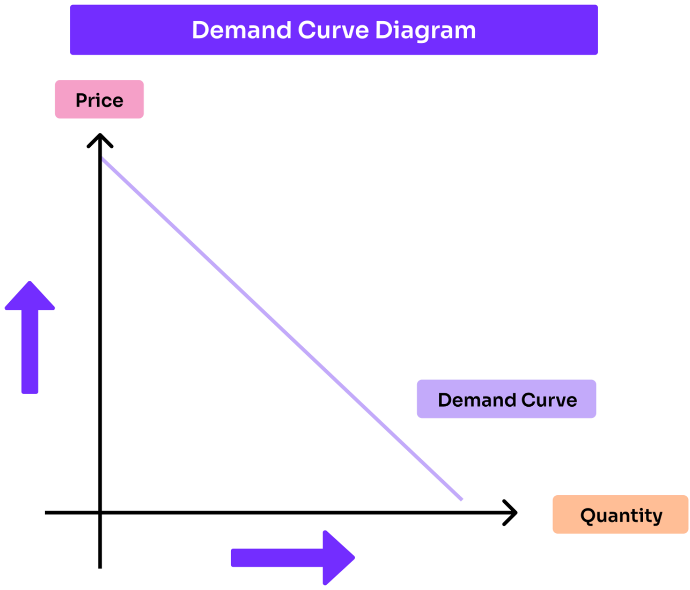


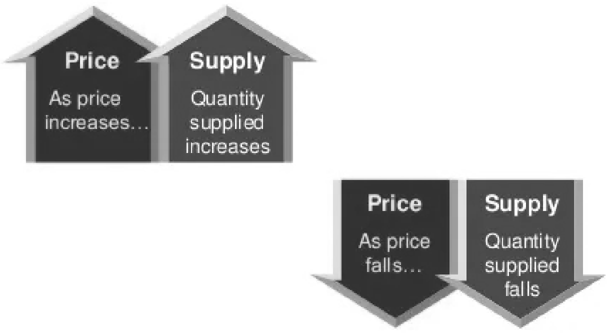
The higher the price of goods, lower will be demand and vice-versa.

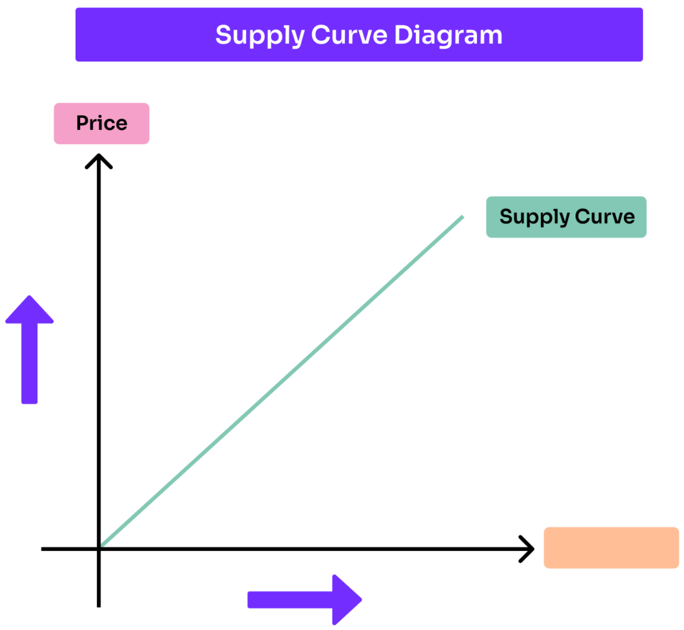
Utility is the level of satisfaction to a consumer by consuming goods or services *For example;* Bread satisfies hunger, TV satisfies the want for entertainment, etc.

Marginal utility refers to the extra utility a consumer gets from one additional unit of a specific product. It is the utility derived by a single unit of consumption.

Supply is defined as the desire with ability to sell and willingness to sell. Supply is a fundamental economic concept that describes the total amount of a specific goods or service that is available to consumers .

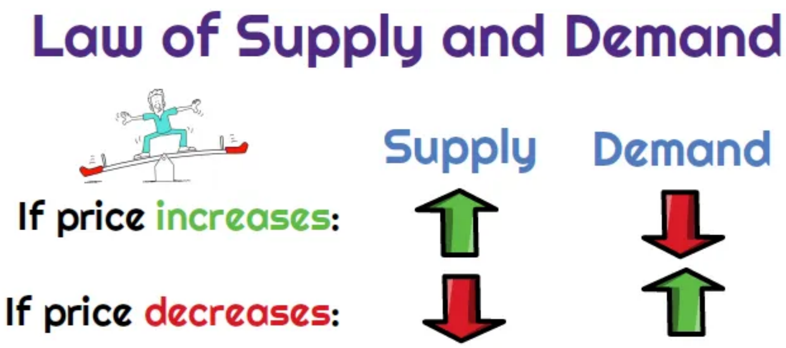
Law of Supply : "All other things being equal, the quantity supplied of a commodity increases when its price increases and vice-versa"**.**





Factors affecting supply

1. New inventions
2. Price of related goods
3. Taxes and subsides
4. Development of infrastructure state of natural resources
5. Change in money income
6. Price of commodity
7. Production technology
8. State of natural resources



Market Type	Main Characteristics	Example

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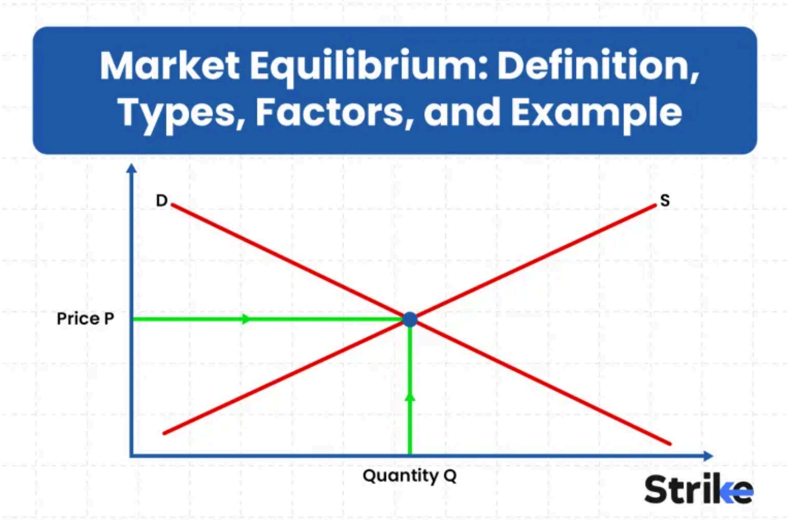
Perfect Competition	Many buyers and sellers; homogeneous products; free entry and exit	Agricultural products such as rice, wheat and vegetables
Monopoly	Single seller; no close substitutes; significant barriers to entry	Electricity transmission/distribution in a particular jurisdiction
Monopolistic Competition	Many sellers differentiated products through brand, quality, design, etc.	Restaurants, clothing and cosmetics
Oligopoly	Few large firms dominate the market; firms are interdependent	Telecommunications and cement industries
Duopoly	Market dominated primarily by two major firms	Visa and Mastercard in global payment networks



Market Equilibrium

The interaction of demand and supply determines the equilibrium price and equilibrium quantity.

A market is in equilibrium when: Qd=Qs

Problem 1: “Supply Creates Its Own Demand” . Justify the Statement

J.B. Say, a French economist of the 19th century, stated that “Supply creates its own demand.” This statement is known as Say’s Law of Markets.

According to Say’s Law, whenever goods and services are produced, the process of production generates income in the form of wages, rent, interest, and profit for the factors of production. This income creates purchasing power, which is then used to purchase goods and services.

Thus, the production of output creates an equivalent purchasing power or demand for goods and services.

Circular Flow of Say’s Law

The basic process can be represented as:

$Output \rightarrow Income \rightarrow Demand \rightarrow Output$

Therefore, according to Say, general overproduction or a deficiency of aggregate demand cannot persist in the long run, because production itself generates the income necessary to purchase the output.

Example

Suppose a factory produces goods worth Rs. 1,00,000.

During production, the factory pays:

- Wages to workers
- Rent to landowners
- Interest to lenders
- Profit to entrepreneurs

These payments generate income of approximately Rs. 1,00,000 for the economy. The recipients of this income can then spend it on goods and services.

Hence:

$Production \rightarrow Income \rightarrow Purchasing\ Power \rightarrow Demand$

This is the basic reasoning behind the statement “Supply creates its own demand.”

Price Mechanism

Another interpretation is based on the price mechanism. If supply increases, prices may fall. Lower prices can encourage consumers to purchase more goods.

$Supply \uparrow \Rightarrow Price \downarrow \Rightarrow Demand \uparrow$

2.2 Elasticity , Application of elasticity and Government Policies (e.g., price ceilings and price floors) (Numerical)

- Elasticity (ε)

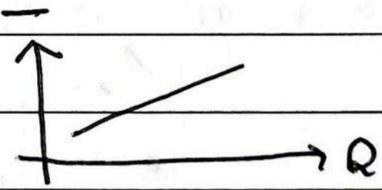
$$\varepsilon = \frac{\text{\% Change in Quantity Demanded}}{\text{\% change in \_\_\_\_\_\_}}$$

$$\varepsilon = \frac{\Delta Q/Q}{\Delta I/I}$$

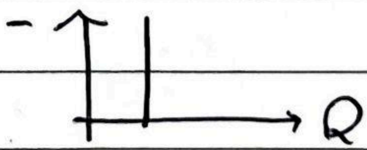
→ Types
- → Price : $\varepsilon_p = \frac{\Delta Q/Q}{\Delta P/P}$
- → Income : $\varepsilon_Y = \frac{\Delta Q/Q}{\Delta Y/Y}$
- → Cross : $\varepsilon_{x,y} = \frac{\Delta Q_x/Q_x}{\Delta P_y/P_y}$

→ Degree of Elasticity :

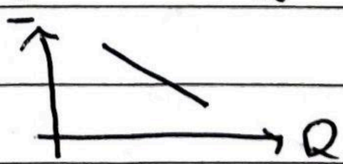
1. Positive Elasticity
($\varepsilon > 0$)



2. Zero Elasticity
($\varepsilon = 0$)



3. Negative Elasticity
($\varepsilon < 0$)



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2.3 Externalities, Market inefficiency (Market Failure)

An externality occurs when an economic activity affects third parties who are not directly involved in the transaction, and this effect is not reflected in market prices.

Types of Externalities

(a) Positive Externalities: Benefits enjoyed by others without payment.

Examples: Education → creates a skilled workforce, Vaccination → reduces spread of disease, Tree plantation → improves air quality

Problem: Market provides less than socially optimal output.

(b) Negative Externalities: Effects imposed on others without compensation.

Examples: Factory pollution, Traffic congestion, Noise from construction sites.

Problem: Market produces more than socially optimal output.

A market inefficiency / market failure occurs when the free market fails to allocate resources efficiently, leading to social welfare loss.

Relationship between Externalities and Inefficiency: Externalities cause divergence between private cost/benefit and social cost/benefit. This divergence leads to deadweight loss and inefficient allocation

Market failure occurs when the free market fails to allocate resources efficiently, leading to loss of social welfare. It is mainly caused due to- externalities, public goods, market power and information asymmetry and so on which result in over- or under-production and that reveals inefficiency.

Policies to overcome market failure	
Tax	Raising price of goods to reduce demand.
Subsidies	Government paying part of the cost
Laws and regulations	Maximum age for buying alcohol
Pollution permits	Firms given right to pollute certain amount
Advertising	Government warning of dangers of tobacco
Nudges	Making it harder to purchase demerit goods
Government price controls	Max prices/min prices / buffer stocks
Changes in property rights	Giving people property rights over river to sue firms who pollute
Policies to reduce unemployment	Education, training, better information
Labour market regulation	Minimum wages, maximum working week www.economicshelp.org

Types of Market Failure	
Public Good	Goods which are non-rival and non-excludable - e.g. police, national defence.
Inequality	Unfair distribution of resources in free market
Monopoly	A firm dominates the market and can set higher prices.
Merit Good	People underestimate the benefit of good, e.g. education.
Factor immobility	E.g. geographical unemployment where people find it difficult to move to areas of jobs.
Agriculture	Volatile prices, fluctuating weather and externalities.
Cyclical instability	Macro economy enters recession or experiences inflationary boom
Externalities	When consuming or producing a good causes a cost/benefit to a third party
Demerit Good	People underestimate the costs of a good, e.g. smoking. www.economicshelp.org
How to remember types of market failure? PIMM FACED!	

2.4 Firm Behavior

Firm behavior refers to how firms decide price, output and production methods to achieve their objectives.

The main objectives are:

1. Profit maximization (MR = MC)
2. Cost minimization
3. Growth
4. Market share

The firm behavior primarily depends on;

i. Market structure (perfect competition, monopoly, oligopoly)

ii. Cost conditions and demand

iii. Government rules and competition

Firm behavior explains how firms make pricing and output decisions to achieve their objectives .

Q. How engineers view the externality due to the economic tradeoff? Explain with examples and curves. With a fall in the price of a commodity from Rs.10 per unit to Rs.8 per unit, the quantity supplied of the commodity falls by 500 units. The price elasticity of supply is 2. Calculate the quantity supplied of this commodity at the price of Rs.8 per unit. [2082 Chaitra]

An externality occurs when the production or consumption of a good affects a third party who is not directly involved in the market transaction. Engineers often evaluate such effects by comparing the private economic benefit/cost with the social benefit/cost.

Negative externality , example: industrial pollution

The basic engineering/economic trade-off is that using resources for one purpose means giving up their alternative use.

$$E_s = \frac{\Delta Q}{\Delta P} \times \frac{P_1}{Q_1}$$

$$2 = \frac{500}{2} \times \frac{10}{Q_1} \Rightarrow Q_1 = 1250$$

$$\therefore Q_2 = 1250 - 500 = \boxed{750}$$